Настройка VPN L2TP+IpSec между VPS CentOS 8 и MikroTik

mikrotiklab.ru/nastrojka_vpn_l2tpipsec_mezhdu_vps_centos_i_mikrotik

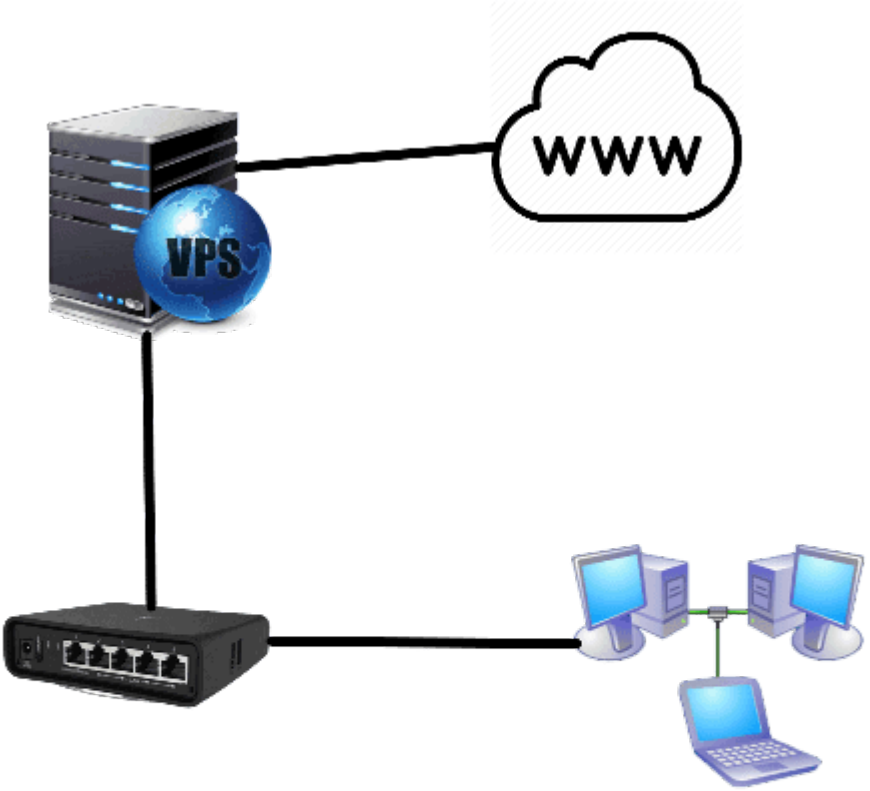
Значительная часть серверной инфраструктуры развёртывается в облаке в формате VPS или VDS, где одним из обязательных условий является обеспечение удалённого доступа через защищённые каналы. Для этой задачи используется VPN-туннель на основе L2TP с поддержкой IPsec.

Задачи, решаемые VPN-туннелем между MikroTik и VPS на CentOS 8

Все запросы, связанные с организацией удалённого доступа к облачным серверам (VPS или VDS), условно делятся на две категории:

- предоставление доступа к локальному серверу;
- маршрутизация трафика через удалённый сервер.

В данной инструкции рассматривается второй вариант — маршрутизация локального трафика через удалённый сервер. Географическое расположение сервера значения не имеет: это может быть США, Германия, Болгария, Россия, Япония и другие страны. Схема процесса выглядит следующим образом:



Настройка VPN L2TP+IPsec между MikroTik и VPS на CentOS 8

Весь комплекс настройки будет разделен на два этапа:

1.Настройка L2TP VPN-сервера на удалённом VPS с CentOS 8

Подключение к удалённому VPS-серверу выполняется по протоколу SSH.

```
yum install nano mc -y
```

Создание скрипта для установки и настройки L2TP+IPsec VPN-сервера

```
nano vpnsetup.sh
```

```
#!/bin/sh
#
# Script for automatic setup of an IPsec VPN server on CentOS/RHEL 6, 7 and 8.
# Works on any dedicated server or virtual private server (VPS) except OpenVZ.
#
# DO NOT RUN THIS SCRIPT ON YOUR PC OR MAC!
#
# The latest version of this script is available at:
# https://github.com/hwdsl2/setup-ipsec-vpn
#
# Copyright (C) 2015-2020 Lin Song <[email protected]>
# Based on the work of Thomas Sarlandie (Copyright 2012)
#
# This work is licensed under the Creative Commons Attribution-ShareAlike 3.0
# Unported License: http://creativecommons.org/licenses/by-sa/3.0/
#
# Attribution required: please include my name in any derivative and let me
# know how you have improved it!

# =====================================================

# Define your own values for these variables
# - IPsec pre-shared key, VPN username and password
# - All values MUST be placed inside 'single quotes'
# - DO NOT use these special characters within values: \ " '

YOUR_IPSEC_PSK=''
YOUR_USERNAME=''
YOUR_PASSWORD=''

# Important notes: https://git.io/vpnnotes
# Setup VPN clients: https://git.io/vpnclients
# IKEv2 guide: https://git.io/ikev2

# =====================================================

export PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
SYS_DT=$(date +%F-%T)

exiterr() { echo "Error: $1" >&2; exit 1; }
exiterr2() { exiterr "'yum install' failed."; }
conf_bk() { /bin/cp -f "$1" "$1.old-$SYS_DT" 2>/dev/null; }
bigecho() { echo; echo "## $1"; echo; }

check_ip() {
IP_REGEX='^(([0-9]|[1-9][0-9]|1[0-9]{2}|2[0-4][0-9]|25[0-5])\.){3}([0-9]|[1-9][0-
9]|1[0-9]{2}|2[0-4][0-9]|25[0-5])$'
printf '%s' "$1" | tr -d '\n' | grep -Eq "$IP_REGEX"
}

vpnsetup() {

if ! grep -qs -e "release 6" -e "release 7" -e "release 8" /etc/redhat-release; then
echo "Error: This script only supports CentOS/RHEL 6, 7 and 8." >&2
echo "For Ubuntu/Debian, use https://git.io/vpnsetup" >&2
exit 1
```

```
fi

if [ -f /proc/user_beancounters ]; then
exiterr "OpenVZ VPS is not supported. Try OpenVPN: github.com/Nyr/openvpn-install"
fi

if [ "$(id -u)" != 0 ]; then
exiterr "Script must be run as root. Try 'sudo sh $0'"
fi

def_iface=$(route 2>/dev/null | grep -m 1 '^default' | grep -o '[^ ]*$')
[ -z "$def_iface" ] && def_iface=$(ip -4 route list 0/0 2>/dev/null | grep -m 1 -Po '(?
<=dev )(\S+)')
def_state=$(cat "/sys/class/net/$def_iface/operstate" 2>/dev/null)
if [ -n "$def_state" ] && [ "$def_state" != "down" ]; then
case "$def_iface" in
wl*)
exiterr "Wireless interface '$def_iface' detected. DO NOT run this script on your PC or
Mac!"
;;
esac
NET_IFACE="$def_iface"
else
eth0_state=$(cat "/sys/class/net/eth0/operstate" 2>/dev/null)
if [ -z "$eth0_state" ] || [ "$eth0_state" = "down" ]; then
exiterr "Could not detect the default network interface."
fi
NET_IFACE=eth0
fi

[ -n "$YOUR_IPSEC_PSK" ] && VPN_IPSEC_PSK="$YOUR_IPSEC_PSK"
[ -n "$YOUR_USERNAME" ] && VPN_USER="$YOUR_USERNAME"
[ -n "$YOUR_PASSWORD" ] && VPN_PASSWORD="$YOUR_PASSWORD"

if [ -z "$VPN_IPSEC_PSK" ] && [ -z "$VPN_USER" ] && [ -z "$VPN_PASSWORD" ]; then
bigecho "VPN credentials not set by user. Generating random PSK and password..."
VPN_IPSEC_PSK=$(LC_CTYPE=C tr -dc 'A-HJ-NPR-Za-km-z2-9' < /dev/urandom | head -c 20)
VPN_USER=vpnuser
VPN_PASSWORD=$(LC_CTYPE=C tr -dc 'A-HJ-NPR-Za-km-z2-9' < /dev/urandom | head -c 16)
fi

if [ -z "$VPN_IPSEC_PSK" ] || [ -z "$VPN_USER" ] || [ -z "$VPN_PASSWORD" ]; then
exiterr "All VPN credentials must be specified. Edit the script and re-enter them."
fi

if printf '%s' "$VPN_IPSEC_PSK $VPN_USER $VPN_PASSWORD" | LC_ALL=C grep -q '[^ -~]\+';
then
exiterr "VPN credentials must not contain non-ASCII characters."
fi

case "$VPN_IPSEC_PSK $VPN_USER $VPN_PASSWORD" in
*[\\\"\']*)
exiterr "VPN credentials must not contain these special characters: \\ \" '"
;;
esac
```

```
bigecho "VPN setup in progress... Please be patient."

# Create and change to working dir
mkdir -p /opt/src
cd /opt/src || exit 1

bigecho "Installing packages required for setup..."

yum -y install wget bind-utils openssl tar \
iptables iproute gawk grep sed net-tools || exiterr2

bigecho "Trying to auto discover IP of this server..."

cat <<'EOF'
In case the script hangs here for more than a few minutes,
press Ctrl-C to abort. Then edit it and manually enter IP.
EOF

# In case auto IP discovery fails, enter server's public IP here.
PUBLIC_IP=${VPN_PUBLIC_IP:-''}

[ -z "$PUBLIC_IP" ] && PUBLIC_IP=$(dig @resolver1.opendns.com -t A -4 myip.opendns.com
+short)

check_ip "$PUBLIC_IP" || PUBLIC_IP=$(wget -t 3 -T 15 -qO- http://ipv4.icanhazip.com)
check_ip "$PUBLIC_IP" || exiterr "Cannot detect this server's public IP. Edit the
script and manually enter it."

bigecho "Adding the EPEL repository..."

epel_url="https://dl.fedoraproject.org/pub/epel/epel-release-latest-$(rpm -E '%
{rhel}').noarch.rpm"
yum -y install epel-release || yum -y install "$epel_url" || exiterr2

bigecho "Installing packages required for the VPN..."

REPO1='--enablerepo=epel'
REPO2='--enablerepo=*server-optional*'
REPO3='--enablerepo=*releases-optional*'
REPO4='--enablerepo=[Pp]ower[Tt]ools'

yum -y install nss-devel nspr-devel pkgconfig pam-devel \
libcap-ng-devel libselinux-devel curl-devel nss-tools \
flex bison gcc make ppp || exiterr2

yum "$REPO1" -y install xl2tpd || exiterr2

if grep -qs "release 6" /etc/redhat-release; then
yum -y remove libevent-devel
yum "$REPO2" "$REPO3" -y install libevent2-devel fipscheck-devel || exiterr2
elif grep -qs "release 7" /etc/redhat-release; then
yum -y install systemd-devel iptables-services || exiterr2
yum "$REPO2" "$REPO3" -y install libevent-devel fipscheck-devel || exiterr2
else
if [ -f /usr/sbin/subscription-manager ]; then
subscription-manager repos --enable "codeready-builder-for-rhel-8-*-rpms"
```

```
yum -y install systemd-devel iptables-services libevent-devel fipscheck-devel ||
exiterr2
else
yum "$REPO4" -y install systemd-devel iptables-services libevent-devel fipscheck-devel
|| exiterr2
fi
fi

bigecho "Installing Fail2Ban to protect SSH..."

yum "$REPO1" -y install fail2ban || exiterr2

bigecho "Compiling and installing Libreswan..."

SWAN_VER=3.32
swan_file="libreswan-$SWAN_VER.tar.gz"
swan_url1="https://github.com/libreswan/libreswan/archive/v$SWAN_VER.tar.gz"
swan_url2="https://download.libreswan.org/$swan_file"
if ! { wget -t 3 -T 30 -nv -O "$swan_file" "$swan_url1" || wget -t 3 -T 30 -nv -O
"$swan_file" "$swan_url2"; }; then
exit 1
fi
/bin/rm -rf "/opt/src/libreswan-$SWAN_VER"
tar xzf "$swan_file" && /bin/rm -f "$swan_file"
cd "libreswan-$SWAN_VER" || exit 1
cat > Makefile.inc.local <<'EOF'
WERROR_CFLAGS = -w
USE_DNSSEC = false
USE_DH2 = true
USE_DH31 = false
USE_NSS_AVA_COPY = true
USE_NSS_IPSEC_PROFILE = false
USE_GLIBC_KERN_FLIP_HEADERS = true
EOF
if ! grep -qs IFLA_XFRM_LINK /usr/include/linux/if_link.h; then
echo "USE_XFRM_INTERFACE_IFLA_HEADER = true" >> Makefile.inc.local
fi
NPROCS=$(grep -c ^processor /proc/cpuinfo)
[ -z "$NPROCS" ] && NPROCS=1
make "-j$((NPROCS+1))" -s base && make -s install-base

cd /opt/src || exit 1
/bin/rm -rf "/opt/src/libreswan-$SWAN_VER"
if ! /usr/local/sbin/ipsec --version 2>/dev/null | grep -qF "$SWAN_VER"; then
exiterr "Libreswan $SWAN_VER failed to build."
fi

bigecho "Creating VPN configuration..."

L2TP_NET=${VPN_L2TP_NET:-'192.168.41.0/24'}
L2TP_LOCAL=${VPN_L2TP_LOCAL:-'192.168.41.1'}
L2TP_POOL=${VPN_L2TP_POOL:-'192.168.41.10-192.168.41.250'}
XAUTH_NET=${VPN_XAUTH_NET:-'192.168.43.0/24'}
XAUTH_POOL=${VPN_XAUTH_POOL:-'192.168.43.10-192.168.43.250'}
DNS_SRV1=${VPN_DNS_SRV1:-'8.8.8.8'}
DNS_SRV2=${VPN_DNS_SRV2:-'8.8.4.4'}
```

```
DNS_SRVS="\"$DNS_SRV1 $DNS_SRV2\""
[ -n "$VPN_DNS_SRV1" ] && [ -z "$VPN_DNS_SRV2" ] && DNS_SRVS="$DNS_SRV1"

# Create IPsec config
conf_bk "/etc/ipsec.conf"
cat > /etc/ipsec.conf <<EOF
version 2.0

config setup
virtual-
private=%v4:10.0.0.0/8,%v4:192.168.0.0/16,%v4:172.16.0.0/12,%v4:!$L2TP_NET,%v4:!$XAUTH_
NET
protostack=netkey
interfaces=%defaultroute
uniqueids=no

conn shared
left=%defaultroute
leftid=$PUBLIC_IP
right=%any
encapsulation=yes
authby=secret
pfs=no
rekey=no
keyingtries=5
dpddelay=30
dpdtimeout=120
dpdaction=clear
ikev2=never
ike=aes256-sha2,aes128-sha2,aes256-sha1,aes128-sha1,aes256-sha2;modp1024,aes128-
sha1;modp1024
phase2alg=aes_gcm-null,aes128-sha1,aes256-sha1,aes256-sha2_512,aes128-sha2,aes256-sha2
sha2-truncbug=no

conn l2tp-psk
auto=add
leftprotoport=17/1701
rightprotoport=17/%any
type=transport
phase2=esp
also=shared

conn xauth-psk
auto=add
leftsubnet=0.0.0.0/0
rightaddresspool=$XAUTH_POOL
modecfgdns=$DNS_SRVS
leftxauthserver=yes
rightxauthclient=yes
leftmodecfgserver=yes
rightmodecfgclient=yes
modecfgpull=yes
xauthby=file
ike-frag=yes
cisco-unity=yes
also=shared
```

```
EOF

# Specify IPsec PSK
conf_bk "/etc/ipsec.secrets"
cat > /etc/ipsec.secrets <<EOF
%any %any : PSK "$VPN_IPSEC_PSK"
EOF

# Create xl2tpd config
conf_bk "/etc/xl2tpd/xl2tpd.conf"
cat > /etc/xl2tpd/xl2tpd.conf <<EOF
[global]
port = 1701

[lns default]
ip range = $L2TP_POOL
local ip = $L2TP_LOCAL
require chap = yes
refuse pap = yes
require authentication = yes
name = l2tpd
pppoptfile = /etc/ppp/options.xl2tpd
length bit = yes
EOF

# Set xl2tpd options
conf_bk "/etc/ppp/options.xl2tpd"
cat > /etc/ppp/options.xl2tpd <<EOF
+mschap-v2
ipcp-accept-local
ipcp-accept-remote
noccp
auth
mtu 1280
mru 1280
proxyarp
lcp-echo-failure 4
lcp-echo-interval 30
connect-delay 5000
ms-dns $DNS_SRV1
EOF

if [ -z "$VPN_DNS_SRV1" ] || [ -n "$VPN_DNS_SRV2" ]; then
cat >> /etc/ppp/options.xl2tpd <<EOF
ms-dns $DNS_SRV2
EOF
fi

# Create VPN credentials
conf_bk "/etc/ppp/chap-secrets"
cat > /etc/ppp/chap-secrets <<EOF
"$VPN_USER" l2tpd "$VPN_PASSWORD" *
EOF

conf_bk "/etc/ipsec.d/passwd"
VPN_PASSWORD_ENC=$(openssl passwd -1 "$VPN_PASSWORD")
```

```
cat > /etc/ipsec.d/passwd <<EOF
$VPN_USER:$VPN_PASSWORD_ENC:xauth-psk
EOF

bigecho "Updating sysctl settings..."

if ! grep -qs "hwdsl2 VPN script" /etc/sysctl.conf; then
conf_bk "/etc/sysctl.conf"
if [ "$(getconf LONG_BIT)" = "64" ]; then
SHM_MAX=68719476736
SHM_ALL=4294967296
else
SHM_MAX=4294967295
SHM_ALL=268435456
fi
cat >> /etc/sysctl.conf <<EOF

# Added by hwdsl2 VPN script
kernel.msgmnb = 65536
kernel.msgmax = 65536
kernel.shmmax = $SHM_MAX
kernel.shmall = $SHM_ALL

net.ipv4.ip_forward = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.all.accept_redirects = 0
net.ipv4.conf.all.send_redirects = 0
net.ipv4.conf.all.rp_filter = 0
net.ipv4.conf.default.accept_source_route = 0
net.ipv4.conf.default.accept_redirects = 0
net.ipv4.conf.default.send_redirects = 0
net.ipv4.conf.default.rp_filter = 0
net.ipv4.conf.$NET_IFACE.send_redirects = 0
net.ipv4.conf.$NET_IFACE.rp_filter = 0

net.core.wmem_max = 12582912
net.core.rmem_max = 12582912
net.ipv4.tcp_rmem = 10240 87380 12582912
net.ipv4.tcp_wmem = 10240 87380 12582912
EOF
fi

bigecho "Updating IPTables rules..."

# Check if rules need updating
ipt_flag=0
IPT_FILE="/etc/sysconfig/iptables"
if ! grep -qs "hwdsl2 VPN script" "$IPT_FILE" \
|| ! iptables -t nat -C POSTROUTING -s "$L2TP_NET" -o "$NET_IFACE" -j MASQUERADE
2>/dev/null \
|| ! iptables -t nat -C POSTROUTING -s "$XAUTH_NET" -o "$NET_IFACE" -m policy --dir out
--pol none -j MASQUERADE 2>/dev/null; then
ipt_flag=1
fi

# Add IPTables rules for VPN
```

```
if [ "$ipt_flag" = "1" ]; then
service fail2ban stop >/dev/null 2>&1
iptables-save > "$IPT_FILE.old-$SYS_DT"
iptables -I INPUT 1 -p udp --dport 1701 -m policy --dir in --pol none -j DROP
iptables -I INPUT 2 -m conntrack --ctstate INVALID -j DROP
iptables -I INPUT 3 -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
iptables -I INPUT 4 -p udp -m multiport --dports 500,4500 -j ACCEPT
iptables -I INPUT 5 -p udp --dport 1701 -m policy --dir in --pol ipsec -j ACCEPT
iptables -I INPUT 6 -p udp --dport 1701 -j DROP
iptables -I FORWARD 1 -m conntrack --ctstate INVALID -j DROP
iptables -I FORWARD 2 -i "$NET_IFACE" -o ppp+ -m conntrack --ctstate
RELATED,ESTABLISHED -j ACCEPT
iptables -I FORWARD 3 -i ppp+ -o "$NET_IFACE" -j ACCEPT
iptables -I FORWARD 4 -i ppp+ -o ppp+ -s "$L2TP_NET" -d "$L2TP_NET" -j ACCEPT
iptables -I FORWARD 5 -i "$NET_IFACE" -d "$XAUTH_NET" -m conntrack --ctstate
RELATED,ESTABLISHED -j ACCEPT
iptables -I FORWARD 6 -s "$XAUTH_NET" -o "$NET_IFACE" -j ACCEPT
# Uncomment if you wish to disallow traffic between VPN clients themselves
# iptables -I FORWARD 2 -i ppp+ -o ppp+ -s "$L2TP_NET" -d "$L2TP_NET" -j DROP
# iptables -I FORWARD 3 -s "$XAUTH_NET" -d "$XAUTH_NET" -j DROP
iptables -A FORWARD -j DROP
iptables -t nat -I POSTROUTING -s "$XAUTH_NET" -o "$NET_IFACE" -m policy --dir out --
pol none -j MASQUERADE
iptables -t nat -I POSTROUTING -s "$L2TP_NET" -o "$NET_IFACE" -j MASQUERADE
echo "# Modified by hwdsl2 VPN script" > "$IPT_FILE"
iptables-save >> "$IPT_FILE"
fi

bigecho "Creating basic Fail2Ban rules..."

if [ ! -f /etc/fail2ban/jail.local ] ; then
cat > /etc/fail2ban/jail.local <<'EOF'
[ssh-iptables]
enabled = true
filter = sshd
action = iptables[name=SSH, port=ssh, protocol=tcp]
logpath = /var/log/secure
EOF
fi

bigecho "Enabling services on boot..."

if grep -qs "release 6" /etc/redhat-release; then
chkconfig iptables on
chkconfig fail2ban on
else
systemctl --now mask firewalld 2>/dev/null
systemctl enable iptables fail2ban 2>/dev/null
fi

if ! grep -qs "hwdsl2 VPN script" /etc/rc.local; then
if [ -f /etc/rc.local ]; then
conf_bk "/etc/rc.local"
else
echo '#!/bin/sh' > /etc/rc.local
fi
```

```
cat >> /etc/rc.local <<'EOF'

# Added by hwdsl2 VPN script
(sleep 15
modprobe -q pppol2tp
service ipsec restart
service xl2tpd restart
echo 1 > /proc/sys/net/ipv4/ip_forward)&
EOF
fi

bigecho "Starting services..."

# Restore SELinux contexts
restorecon /etc/ipsec.d/*db 2>/dev/null
restorecon /usr/local/sbin -Rv 2>/dev/null
restorecon /usr/local/libexec/ipsec -Rv 2>/dev/null

# Reload sysctl.conf
sysctl -e -q -p

# Update file attributes
chmod +x /etc/rc.local
chmod 600 /etc/ipsec.secrets* /etc/ppp/chap-secrets* /etc/ipsec.d/passwd*

# Apply new IPTables rules
iptables-restore < "$IPT_FILE"

# Fix xl2tpd on CentOS 7/8, if kernel module "l2tp_ppp" is unavailable
if grep -qs -e "release 7" -e "release 8" /etc/redhat-release; then
if ! modprobe -q l2tp_ppp; then
sed -i '/^ExecStartPre/s/^/#/' /usr/lib/systemd/system/xl2tpd.service
systemctl daemon-reload
fi
fi

# Restart services
mkdir -p /run/pluto
modprobe -q pppol2tp
service fail2ban restart 2>/dev/null
service ipsec restart 2>/dev/null
service xl2tpd restart 2>/dev/null

cat <<EOF

================================================

IPsec VPN server is now ready for use!

Connect to your new VPN with these details:

Server IP: $PUBLIC_IP
IPsec PSK: $VPN_IPSEC_PSK
Username: $VPN_USER
Password: $VPN_PASSWORD
```

```
Write these down. You'll need them to connect!

Important notes: https://git.io/vpnnotes
Setup VPN clients: https://git.io/vpnclients
IKEv2 guide: https://git.io/ikev2

================================================

EOF

}

## Defer setup until we have the complete script
vpnsetup "$@"

exit 0
```

При необходимости изменения параметров сетевой адресации следует использовать данный раздел.

```
L2TP_NET=${VPN_L2TP_NET:-'192.168.41.0/24'}
L2TP_LOCAL=${VPN_L2TP_LOCAL:-'192.168.41.1'}
L2TP_POOL=${VPN_L2TP_POOL:-'192.168.41.10-192.168.41.250'}
```

выполнение скрипта

```
sh vpnsetup.sh
```

После завершения выполнения скрипт отображает параметры, которые необходимо указать в настройках L2TP-клиента на MikroTik.

```
Server IP: 18.13.12.14
IPsec PSK: YsH8f7ke3r6i6TBB5pZu
Username: vpnuser
Password: 2FG2i3aTHm3b7soo
```

Для обеспечения работы интернета в локальной сети за MikroTik требуется внести изменения в **IPTABLES**.

```
nano /etc/sysconfig/iptables

-A POSTROUTING -s 192.168.10.0/24 -o ens3 -j MASQUERADE
-A FORWARD -s 192.168.10.0/24 -d 192.168.10.0/24 -i ppp+ -o ppp+ -j ACCEPT

systemctl restart iptables
```

Файл xl2tpd.conf содержит настройки VPN сервера:

```
nano /etc/xl2tpd/xl2tpd.conf
```

```
[global]
port = 1701

[lns default]
ip range = 192.168.41.10-192.168.41.250
local ip = 192.168.41.1
require chap = yes
refuse pap = yes
require authentication = yes
name = l2tpd
pppoptfile = /etc/ppp/options.xl2tpd
length bit = yes
```

Необходимо добавить маршрут в подсеть, расположенную за MikroTik. Данное правило будет автоматически применяться при каждом поднятии туннеля.

```
nano /etc/ppp/ip-up

ip route add 192.168.10.0/24 via 192.168.41.10
```

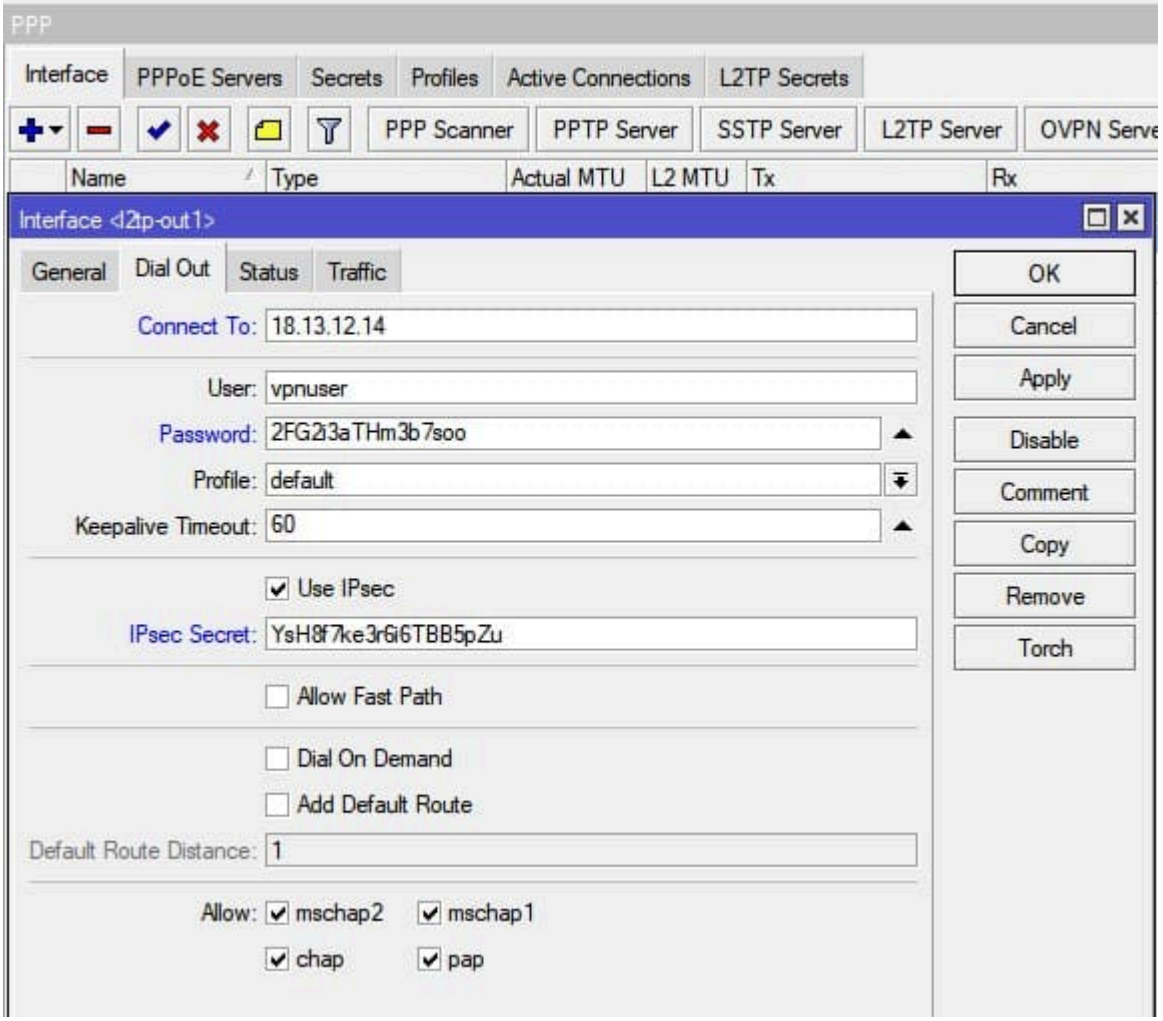
Здесь:

- **192.168.10.0/24** — подсеть, расположенная за MikroTik.
- **192.168.41.10** — адрес MikroTik на стороне сервера CentOS 8.

2.Настройка L2TP клиента на MikroTik

На стороне MikroTik необходимо настроить два элемента:

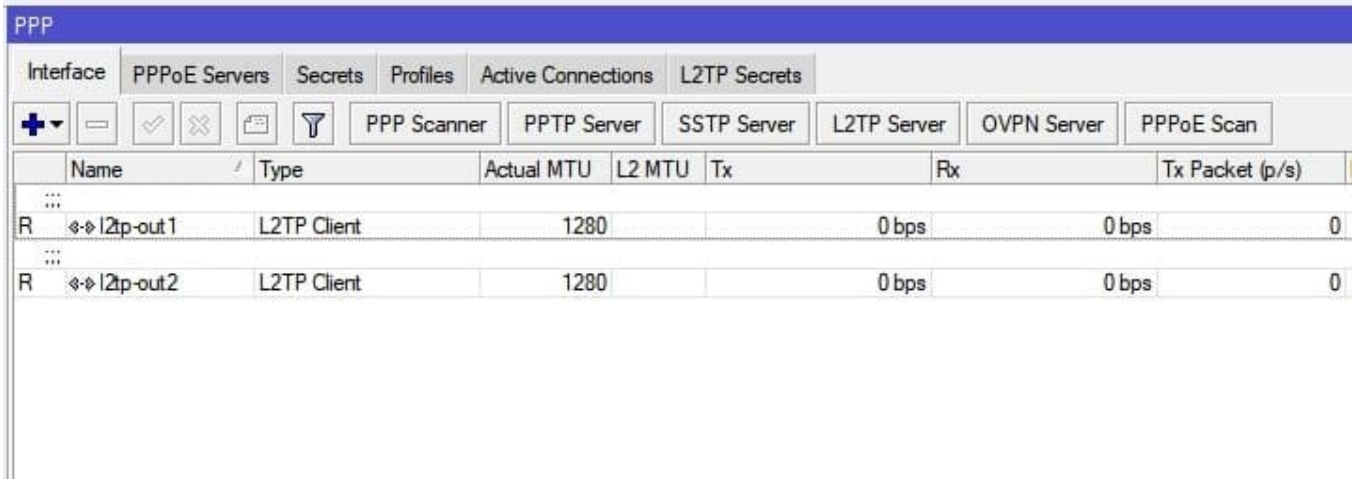
1. **PPP-клиент** — создание L2TP-подключения. Настройка выполняется в меню **PPP** → **Interface**.
2. **Маршрутизация** — добавление маршрутов для трафика через созданный туннель.



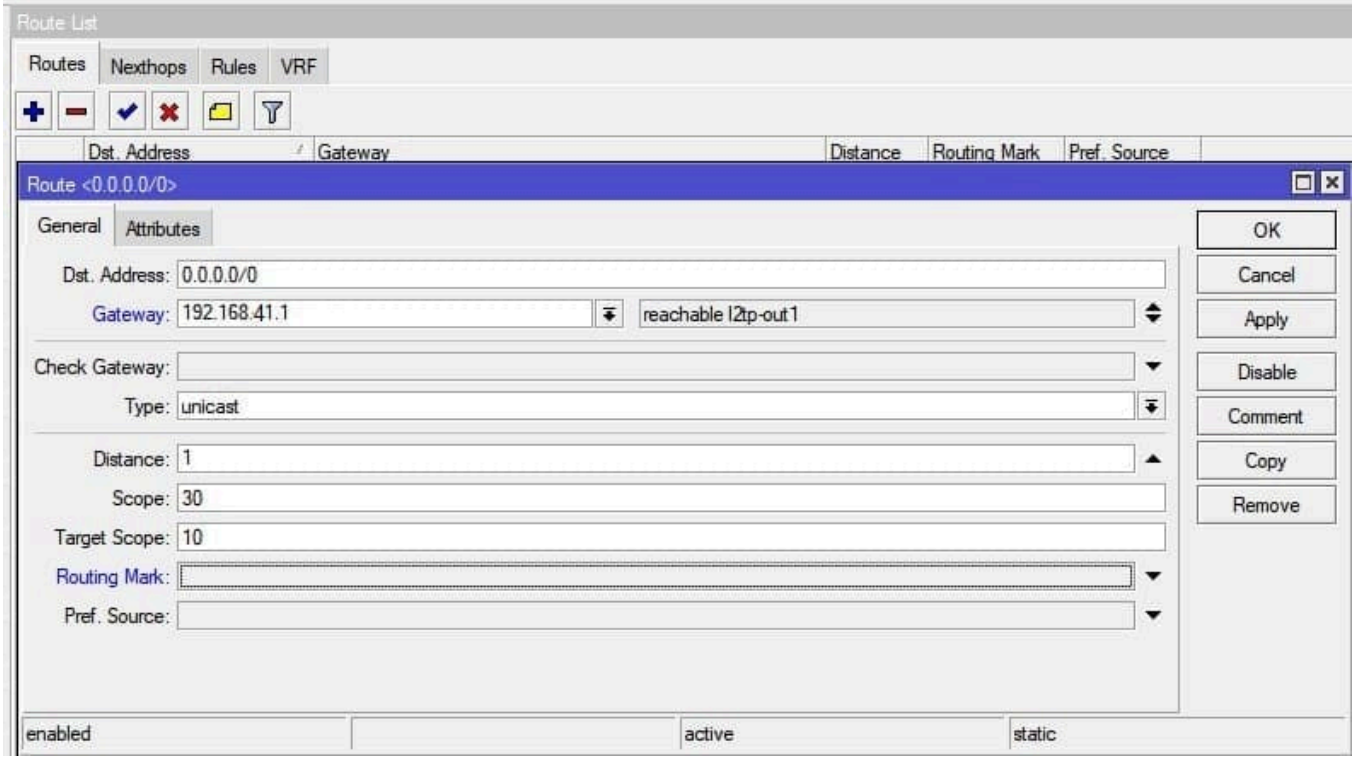
ОНЛАЙН-КУРС С ПОДДЕРЖКОЙ КУРАТОРА
УЗНАТЬ БОЛЬШЕ
mikrotik под контролем
ЧЕТКАЯ СТРУКТУРА + АКТУАЛЬНЫЕ РЕШЕНИЯ + ПРАКТИЧЕСКИЙ ПОДХОД

```
/ppp profile
/interface l2tp-client
add comment=VPS-CentOS connect-to=18.13.12.14 disabled=no ipsec-secret=\
YsH8f7ke3r6i6TBB5pZu name=l2tp-out1 password=2FG2i3aTHm3b7soo profile=\
default use-ipsec=yes user=vpnuser
```

Успешное соединение, сопровождается статусом «R»

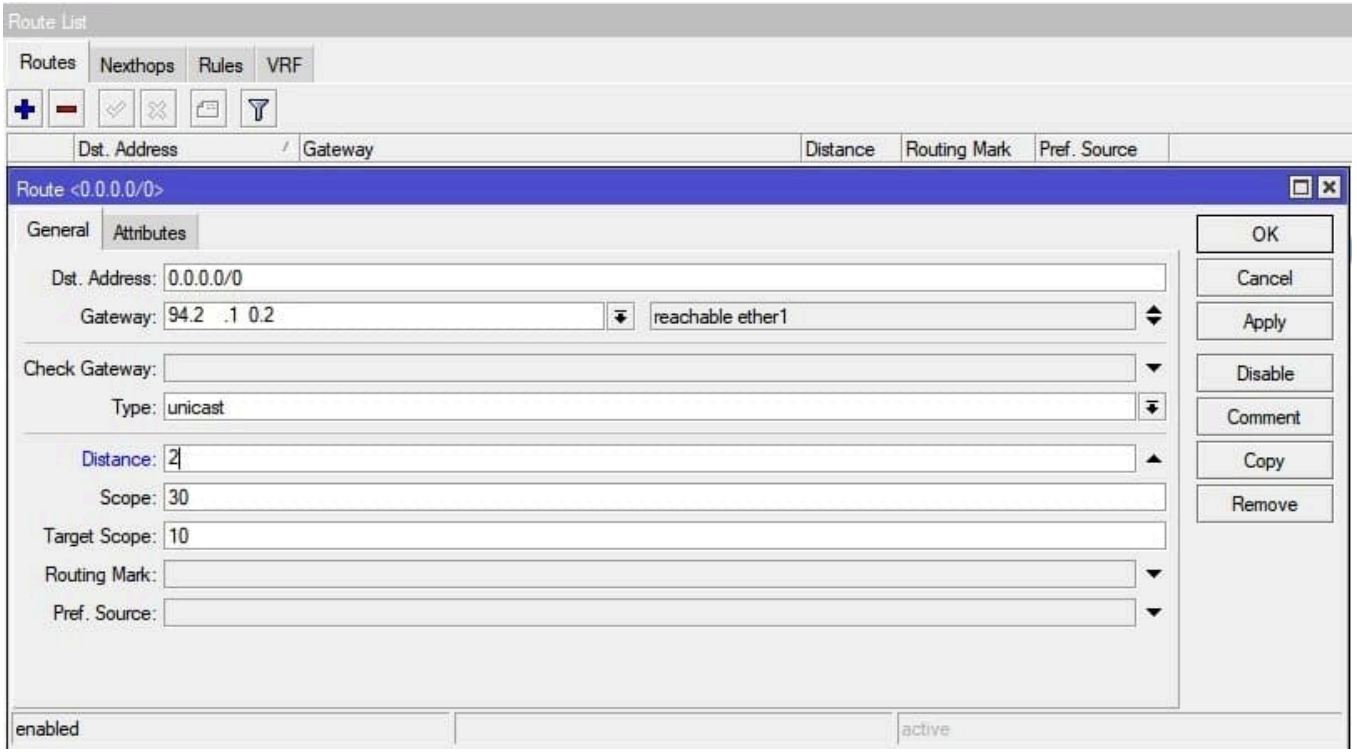


Необходимо добавить статический маршрут, чтобы весь интернет-трафик направлялся через VPN-туннель на CentOS VPS. Конфигурация выполняется в меню **IP** → **Routes**.




```
/ip route
add distance=1 gateway=192.168.41.1
```

А у прежнего интернет маршрута, необходимо понизить **Distance** до 2




```
/ip route
add distance=2 gateway=94.2.XX.YY
```

В результате весь интернет-трафик будет проходить через удалённый VPS. При отсутствии соединения с сервером маршрутизация автоматически переключится на основного интернет-провайдера.